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Statistical Terminology Termbank Report

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Abstract

This report documents the design, implementation and evaluation of a Chinese English bilingual statistical termbase. The termbase takes ISO 3534 and its Chinese counterpart standard GB/T 3358 as its basis, includes 40 vetted terms, covers 37 fields, distributed across 8 relational tables, and supports 17 SQL queries, which are organised by four categories of retrieval dimensions. A matching set of HTML based interactive visualisation scripts provides dynamic demonstrations for several core concepts, and is incorporated into the database through a dedicated demonstration resource field and query. This project sits at the intersection of computational lexicography and relational database design, and its core claim is that the normative conceptual system defined by ISO can be systematically transformed into queryable, extensible lexicographic infrastructure. This report explains the methodological decisions behind term selection, attribute architecture, data collection and database implementation, and concludes with an evaluation.

1. Introduction

Statistical terms have long posed challenges for teaching, translation and cross disciplinary communication. Unlike ordinary vocabulary, statistical terms operate within a tightly delimited conceptual system, in which subtle differences produce practical consequences, for example the difference between population and sample, the difference between accuracy and precision, the difference between fixed effect and random effect. Existing reference resources can only partially meet these needs. Standard dictionaries lack formalised definitions and inter term relations, statistical textbooks often embed terms within procedural descriptions, and standard documents such as ISO 3534 (2006a, 2006b, 2013, 2023), although complete in content, are not designed for interactive retrieval or bilingual cross referencing.

This project establishes a termbase precisely in response to this gap. The entries in the termbase are organised around concepts rather than word forms, and inter term relations are also presented as first class data objects, rather than hidden within explanatory text. The distinction between a termbase and a general dictionary can be understood as follows. A dictionary puts words first; one first locates the headword, and then retrieves its meaning. A termbase puts concepts first; one first represents a concept, and then connects it to its corresponding lexical realisations, related concepts, formal symbols and contextual usages (Hartmann & James, 1998; Sager, 1990).

ISO 3534 provides the normative basis for general statistical concepts, namely Part 1, applied statistics, namely Part 2, experimental design, namely Part 3, and sampling surveys, namely Part 4. This standard provides a network of conceptual relations, namely a network composed of hierarchical relations, associative relations and equivalence relations, allowing statistical knowledge to maintain internal coherence. GB/T 3358 (2009a, 2009b, 2009c) is the Chinese national standard that directly corresponds to ISO 3534; it extends this network into the bilingual domain, providing Chinese terms that are conceptually equivalent, rather than merely literal translations.

Encoding this normative conceptual system into a relational database schema is the main contribution of this project, making the semantic relations between terms a structure that can be queried by computer. The termbase further integrates usage data drawn from corpora, including collocations and example sentences, and incorporates a dedicated set of interactive visualisation materials, thereby becoming a multi purpose resource that simultaneously serves learners, translators and researchers in the language sciences.

2. Term Selection

The construction of any termbase begins with a principled judgement about headword selection,

namely which terms should be included, and why. Frequency driven selection is suitable for general purpose learner dictionaries, but is not suitable for a domain termbase, because the latter places more value on coverage of the conceptual system, rather than coverage of the most common words. This project adopts a workflow coverage criterion, namely that the selected terms should be able to present the complete conceptual stock required for carrying out a quantitative linguistics study.

2.1 The Five-Stage Workflow Framework

The selection criterion is composed of five stages. Each stage contributes 8 terms to the 40 term wordlist, thereby forming a balanced distribution across the workflow stages.

Stage	English Label	Conceptual Domain
A	Define objectives	Probability theory, distributional concepts, parameter types
B	Design study	Sampling design, experimental structure, power analysis
C	Collect data	Annotation reliability, data collection procedures
D	Analyse and infer	Descriptive statistics, effect size, inferential procedures
E	Statistical modelling	Regression, correlation, mixed models, resampling

This deliberately arranged symmetry guarantees that no single methodological stage occupies an excessive share of the resource, but at the cost that conceptual density may not be entirely balanced. Stage A concepts are relatively foundational, and so the relations among terms tend to be richer. Stage E concepts are more methodologically specialised.

2.2 Standards-Driven and Extended Terms

Term selection is primarily sourced from ISO 3534 (2006a, 2006b, 2013, 2023), supplemented by relevant entries from the Berkeley Statistics Glossary that are oriented towards linguistics and the language sciences. The former provides terms with normative definitions, embedded within the conceptual network of the standard. Extended terms are also incorporated, in particular advanced modelling techniques such as Generalised Linear Mixed Models and the Bootstrap, because they embody current methodological practice in language science research. If these terms were excluded, conceptual gaps would appear in the coverage of the research workflow.

2.3 Distribution Across Subcategories

Subcategory	Term Count	Notes
Foundational Statistics	10	Probability, distributions, parameters
Experimental Design	9	Factors, blocking, replication, power
Statistical Modelling	8	Regression, correlation, mixed models
Survey Sampling	5	Frames, designs, sampling error
Annotation Methods	3	Agreement coefficients

Subcategory	Term Count	Notes
Applied Statistics	3	Cross-disciplinary applications
Statistical Inference	2	Integrated into modelling terms at Stage E

Stage A terms, such as Population, Random Variable, Normal Distribution, Variance, Expectation, are mainly ontological in character, because they define the entities and structures within the statistical world. Stage E terms, such as Linear Regression, Bootstrap, Bayesian Inference, Linear Mixed effects Model, are mainly methodological in character, because they define procedures applied to these entities. This gradient has practical implications for attribute design; relevant content is found in Section 3.

3. Attributions

We have attempted to make the termbase capable of representing a concept from multiple simultaneously existing dimensions. The 37 fields of this termbase are organised into four functional groups.

3.1 Four-Dimensional Field Architecture

Field Group	Fields Included	Lexicographic Dimension
Linguistic & Identification	中文名称, 英文名称, 拼音, 词性, 缩写符, 编号 ID	Form dimension — formal and identification properties
Semantic & Relational	中文定义, 英文定义, COBUILD 定义, 词源, 同义词, 反义词, 易混淆概念, 相关概念	Semantic + relational dimensions — meaning and interterm connections
Contextual & Usage	术语搭配, 例句 Prose, 例句来源, 应用场景, 常用平台, 适用于可视化, 演示资源	Pragmatic dimension — attested usage and visualization resources
Technical & Administrative	公式, 变量, 取值范围, 使用条件, 功能描述, 缺点, 来源, 审核人, 审核日期	Formal + traceability dimensions — mathematical specs and quality control

Here, synonyms, antonyms, easily confused concepts and related concepts are precisely where termbase entries differ from dictionary entries. They encode the position of a term within the conceptual network, making the connections among terms a computable object, rather than mere implicit information within explanatory text.

3.2 Formula

A particular challenge faced by a statistical termbase is the representation of mathematical notation. In the current implementation, the formula for standard deviation $\sigma = \sqrt{[\Sigma(x_i - \mu)^2 / N]}$ and the formula for Cohen's Kappa $\kappa = (P_o - P_e) / (1 - P_e)$, and so on, are stored as strings in the "formula" field. This allows substring matching by symbol name, for example searching for σ or β , but cannot

support mathematical reasoning over the structure of the formula.

3.3 Normalization

Not all 37 fields can be assigned a single value for every term. Collocation, example sentence and lexical relation entries may all logically have multiple values. In the current implementation, multi valued fields are stored as separated strings within a single database cell. This is a simplification at the level of First Normal Form, exchanging analytical flexibility for implementational simplicity. A fully normalised schema would represent every collocation, every synonym pair and every easily confused term relation as a row in a dedicated junction table, thereby supporting independent retrieval, counting and relation type annotation (Sager, 1990).

4. Data Collection

37 fields multiplied by 40 (terms) constitute 1,480 discrete data points. Populating these reliably requires a workflow with hierarchical control.

4.1 Collection Workflow

First, definitions are extracted from ISO 3534 and GB/T 3358. Where definitional asymmetry exists between the two standards, accounting for approximately 15% to 20% of the total entries, the English definition field follows ISO, the Chinese definition field follows GB/T, and the discrepancy is flagged in the administrative field. In the resource review section, etymological data is taken from the Oxford English Dictionary. Discipline specific application contexts come from methodological handbooks and textbooks. Subsequently, collocations and example sentences are taken from English academic usage in PubMed and Chinese academic usage in CNKI respectively. This corpus based approach grounds these fields in genuinely attestable disciplinary usage, rather than artificially fabricated example sentences. At the same time, the formula, variable, value range and platform fields are populated according to authoritative textbooks and the official documentation of R and Python statistical packages. All information adopts dual source verification. Finally, all entries are validated by two reviewers. Where opinions diverge, resolution is achieved through structured arbitration with reference to the primary standard sources.

4.2 Sources

Field Type	Primary Source	Secondary Source	Notes
Definitions	ISO 3534 / GB/T 3358	Berkeley Glossary	ISO takes precedence; GB/T for Chinese field
Etymology	Oxford English Dictionary	etymonline.com	Single authority for English word history
Collocations	PubMed / CNKI corpora	—	Corpus-driven, attested disciplinary usage
Example sentences	Corpus extraction	—	Genre: empirical research articles

Field Type	Primary Source	Secondary Source	Notes
Formulas & variables	Authoritative textbooks	R/Python documentation	Dual-source verification required
Application context	Methodological handbooks	Textbooks	Stage-appropriate scope

4.3 Ordinal Logic

Definitions must first be established in Stage 1, before corpus extraction can proceed in Stage 3, because the semantic boundaries of a term determine which collocational instances belong to relevant usage, and which are merely semantic drift or homography. Likewise, the technical population in Stage 4 also requires that definitions and application contexts be first established, namely Stages 1 to 2, before formal attributes can be accurately specified. The cross validation stage is placed last, because it requires complete entries in order to evaluate internal consistency.

5. Database Implementation

5.1 Relational Schema

The database of this project is composed of 8 tables and 1 view. [Full Termbase] aggregates all fields for end user querying, and embodies the normalisation decisions. The terms table stores the primary identifier and descriptive fields, while the secondary tables handle multi valued data or data that is structurally relatively independent, including inter term relations, collocations, technical attributes, visualisation resources and administrative metadata. The foreign key structure ensures referential integrity, namely that every lexical relation, collocation, technical attribute and visualisation resource is traceable to a specific term entry.

```
-- Entity
terms (
  term_id      PK,
  id_label     TEXT,      -- e.g. S01, S37
  term_zh      TEXT,
  term_en      TEXT,
  pinyin       TEXT,
  pos          TEXT,
  symbol       TEXT,
  definition_zh TEXT,
  definition_en TEXT,
  definition_cobuild TEXT,
  workflow_stage CHAR(1), -- A / B / C / D / E
  subcategory  TEXT,
  discipline   TEXT
)
```

```

-- Interterm semantic relations
lexical_relations (
  relation_id PK,
  term_id_from FK → terms.term_id,
  term_id_to FK → terms.term_id,
  relation_type TEXT          -- synonym / antonym / confusable / related
)

-- Corpus-derived collocations
collocations (
  colloc_id PK,
  term_id FK → terms.term_id,
  collocate TEXT,
  example_prose TEXT,
  source_citation TEXT
)

-- Mathematical specifications
technical (
  tech_id PK,
  term_id FK → terms.term_id,
  formula_latex TEXT,
  variables TEXT,
  value_range TEXT,
  conditions TEXT,
  platforms TEXT
)

-- Visualization resources
visualizations (
  viz_id PK,
  term_id FK → terms.term_id,
  resource_label TEXT,
  resource_type TEXT,          -- local / web
  suitability_rating SMALLINT  -- 1–3 stars
)

```

5.2 Visualization Resources

A key feature of this database is the integration of interactive visualisation materials. The project contains a portal index page listing all 40 terms and their associated demonstration resources, and also contains 4 standalone HTML animation files that can be hosted locally. Each animation can run self sufficiently, fully supports offline use, and is designed around a linguistic research scenario.

Animation File	Term(s) Covered	Stage	Conceptual Focus
Linear Mixed-effects Model	S37 — Linear Mixed-effects Model	E	Fixed vs. random effects; shrinkage (Partial Pooling)
Cohen's Kappa	S18 — Cohen's Kappa	C	Contingency table construction;

Animation File	Term(s) Covered	Stage	Conceptual Focus
			Kappa formula derivation
Interaction Effect	S23 — Interaction Effect	C	2×2 interaction plot; ordinal vs. disordinal interactions
BAP (Bias, Accuracy, Precision)	S27–S29 — Bias, Accuracy, Precision	D	Target diagram; sampling distribution linkage; slider controls

In addition to these 4 local animations, the portal also links curated external interactive resources for 23 of the 40 terms, covering probability distributions, regression, correlation, power analysis, Bootstrap resampling and Bayesian inference, and so on. The "demonstration resource" field stores direct references to these materials, and query Q17, namely Q_COBUILD_has_demonstration_resource, retrieves all entries with visualisation resources, and ranks them by applicability score. This integration means that the database is not only a reference resource, but also becomes a navigation layer for a broader pedagogical media ecology.

5.3 Query Design and Access Dimensions

The 17 implemented SQL queries are organised by four retrieval dimensions, corresponding to four categories of differing user orientations.

COBUILD-style Lexicographic Retrieval (Queries Q16–Q17)

The COBUILD retrieval model inverts the lookup direction of traditional dictionaries, allowing users to search starting from usage. For example, a user can search for a particular collocation, example sentence or visualisation tag, and thereby find the corresponding entry. This corpus first orientation embodies the epistemological stance of corpus lexicography (Sinclair, 1991).

```
-- Q_COBUILD_全部词条: All entries in COBUILD lexicographic order
-- headword → POS → definition → example → collocations → pragmatics → media
SELECT 英文名称, 中文名称, 词性, 缩写符,
       英文定义, 中文定义, [例句 Prose], 例句来源,
       术语搭配, 应用场景, 易混淆概念, 相关概念,
       演示资源, 适用于可视化, 子类别, [5 阶段]
FROM [Full Termbase]
ORDER BY [5 阶段], 编号 ID;
```

The COBUILD parameterised search, namely Q17, simultaneously performs wildcard matching across the English name, Chinese name, prose example sentence and collocation fields, thereby supporting entry discovery via fragmentary input or cross field keyword input. The visualisation filter variant, namely Supp_Q1, retrieves entries ranked by applicability score, allowing the database to be used as a navigation interface for animation resources.

Technical and Formal Retrieval (Queries Q11–Q12)

Technical retrieval is oriented towards users requiring formalised mathematical descriptions. The symbol reverse lookup query, namely Q12, provides a usage that is uncommon in standard dictionaries. The user enters a mathematical symbol, and can thereby retrieve the corresponding term.


```
-- Q_公式_按符号查找: Symbol reverse-lookup
-- User supplies symbol ( $\sigma$ ,  $\mu$ ,  $\chi$ ,  $\beta$ ,  $\kappa$ ); retrieves matching terms
SELECT 中文名称, 英文名称, 缩写符, 公式, 变量, 中文定义
FROM [Full Termbase]
WHERE (公式 Like '*' & [symbol_input] & '*')
      OR (变量 Like '*' & [symbol_input] & '*')
ORDER BY 子类别, 编号 ID;
```

Linguistic and Etymological Retrieval (Queries Q9–Q10, Q13–Q15)

Etymological and lexical comparison queries serve those users who focus on the linguistic attributes rather than the mathematical attributes of statistical terms. Understanding that the word statistic originates from the Latin statisticum, meaning relating to the state, or that the word parameter is composed of the Greek para, meaning beside, and metron, meaning measure, helps learners build more stable mental models.

```
-- Q_词源_含特定词根: Etymology root search
-- Accepts Latin/Greek roots or eponymous names (Fisher, Pearson, Cohen, Bayes, Galton)
SELECT 词源, 中文名称, 英文名称, 中文定义, 子类别, [5 阶段]
FROM [Full Termbase]
WHERE 词源 Like '*' & [root_input] & '*'
ORDER BY 词源, 编号 ID;
```

The discrimination queries, namely Q13 and Q14, retrieve easily confused term pairs, for example accuracy and precision, variance and standard deviation, fixed effect and random effect, thereby supporting contrastive terminological analysis. Full text keyword search, namely Q15, simultaneously matches Chinese name, English name, Chinese definition and related concepts.

Workflow-Structured Learning Path Retrieval (Queries Q1–Q5)

Each of the five workflow stages has one query. The Stage D and Stage E queries incorporate "formula" and "variable" fields, while these fields do not exist in the Stage A to C queries, which reflects the increased degree of mathematical formalisation in the later stages. Stage E also retrieves value ranges and conditions of use, in order to support fine grained applied judgement among various modelling options.

```
-- Q_阶段_统计建模: Stage E — Statistical Modelling
-- Adds 公式 and 变量 fields absent from Stage A–C queries
SELECT 中文名称, 英文名称, 子类别, 缩写符,
      中文定义, 英文定义, 应用场景, 公式, 变量,
      常用平台, 演示资源, 词性
FROM [Full Termbase]
WHERE [5 阶段] = '统计建模'
ORDER BY 子类别, 编号 ID;
```

This stage based query design operationalises progressive disclosure into concrete operations. Users first encounter conceptual vocabulary and distributional vocabulary, before entering formalised

mathematical descriptions, which aligns with a principled pedagogical sequence.

5.4 Dictionary Styles Supporting

The termbase simultaneously supports two entry presentation modes. The following table demonstrates both, taking standard deviation as an example.

Dimension	Formal Definition	COBUILD style
Definition starting point	Mathematical formula: $\sigma = \sqrt{[\sum(x_i - \mu)^2 / N]}$	Usage context: 'Standard deviation describes how spread out values are around the mean'
Target user	Expert researcher	Learner / cross-disciplinary reader
Primary database fields	definition_en, 公式, 变量	definition_cobuild, 例句 Prose, 术语搭配
Canonical query	Q_公式_含公式术语 (Q11)	Q_COBUILD_参数检索 (Q17)
Discovery direction	Symbol $\rightarrow$ term name	Usage / example $\rightarrow$ term

This design enables the same underlying data to serve both researchers with statistical training and graduate students in the language sciences who are encountering quantitative methods for the first time. The difference between the two is reflected only in the query method adopted and the presentation layer.

6. Evaluation

6.1 Strengths

Taking ISO 3534 and GB/T 3358 as the definitional basis grounds the termbase in a conceptual system, rather than an ad hoc cobbled together vocabulary list. The hierarchical structure and associative structure of ISO 3534 mean that every entry participates in a relational network sanctioned by the standard, which is a property that no textbook glossary can replicate.

The 37 field supports five categories of access modes of differing nature, namely COBUILD style access, technical access, etymological access, contrastive access and workflow organised access. Each additional layer of field depth expands the queryable surface, while also increasing the editorial cost.

Within the "demonstration resource" field, the applicability scoring system provides a structured navigation interface. The 4 local HTML animations are embedded in language science contexts, enhancing pedagogical specificity.

The 37 inter term links stored in the database, distributed across 27 relation types, constitute a prototype graph. Each relation record is a semantic triple of the form term₁, relation_type, term₂, which can be directly mapped to RDF triples. Therefore, the current relational schema is architecturally already prepared for migration to a knowledge graph representation, for example RDF or SKOS, and can thereby support inferencing.

6.2 Limitations

Scale

40 terms is not yet sufficient to constitute a semantically closed conceptual network. Even within ISO 3534 alone, the total number of defined terms exceeds 200. At the current scale, many terms only connect to one or two other terms, thereby producing conceptual islands that cannot be reached by traversal along relations from a starting term. Claims about the coverage of the network of statistical knowledge must therefore be qualified. What this termbase covers is a representative sample of the conceptual space, not a comprehensive mapping of it.

Formula Storage as String Data

Representing formulas as strings facilitates reading, but cannot support mathematical reasoning over the structure. Symbol search relies on exact character matching.

Single-User Infrastructure

This database platform does not support concurrent multi user access, nor collaborative editing under version control, nor REST API integration, which affects whether the termbase can be maintained long term by other teams. Shareable data standards such as TBX, namely TermBase eXchange, ISO 30042 (2019), and SKOS, namely Simple Knowledge Organization System, W3C (2009), can help overcome these limitations.

7. Conclusion

This project demonstrates the feasibility of encoding the ISO 3534 concept structure into a relational termbank. By representing definitions, bilingual equivalents, semantic relations, formulas, corpus examples, and visualization resources as database fields and linked tables, the prototype makes selected statistical terminology searchable through multiple retrieval paths. The current version contains 40 bilingual entries and 17 SQL queries, and it is usable as a prototype for learners, translators, and researchers in the language sciences. Further work should expand coverage, normalize multivalued attributes, improve formula representation, and support collaborative maintenance through interoperable formats such as TBX or SKOS.

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Appendix A: SQL Query Reference

All 17 implemented SQL queries are described in the table below. Queries are organized by retrieval dimension (5-Stage Learning Path, Disciplinary, Etymology, Formula/Symbol, Lexical Contrast, COBUILD-style). For each query, the table lists its identifier, category, Chinese name, functional purpose, fields retrieved, filter condition, and sort order.

Q ID	Category	Chinese Name	English Description	Fields Selected	Filter Condition	Sort Order
Q01	5-Stage Learning Path	Q_阶段_明确目标	Stage A – Define Objectives: retrieves all terms at the conceptual-definition stage with names, definitions, application context, platforms, and visualization resources	中文名称, 英文名称, 子类别, 缩写符, 中文定义, 英文定义, 应用场景, 常用平台, 演示资源, 词性	[5 阶段] = '明确目标'	子类别, 编号 ID
Q02	5-Stage Learning Path	Q_阶段_设计方案	Stage B – Design Study: retrieves sampling design, experimental structure, and power analysis terms	中文名称, 英文名称, 子类别, 缩写符, 中文定义, 英文定义, 应用场景, 常用平台, 演示资源, 词性	[5 阶段] = '设计方案'	子类别, 编号 ID
Q03	5-Stage Learning Path	Q_阶段_实施采集	Stage C – Collect Data: retrieves annotation reliability and data collection terms; same field set as Stages A–B	中文名称, 英文名称, 子类别, 缩写符, 中文定义, 英文定义, 应用场景, 常用平台, 演示资源, 词性	[5 阶段] = '实施采集'	子类别, 编号 ID
Q04	5-Stage Learning Path	Q_阶段_统计建模	Stage E – Statistical Modelling: adds 公式 and 变量 fields absent from earlier stages, reflecting increased mathematical formality	中文名称, 英文名称, 子类别, 缩写符, 中文定义, 英文定义, 应用场景, 公式, 变量, 常用平台, 演示资源, 词性	[5 阶段] = '统计建模'	子类别, 编号 ID

Q ID	Category	Chinese Name	English Description	Fields Selected	Filter Condition	Sort Order
Q05	5-Stage Learning Path	Q_阶段_分析推断	Stage D – Analyse and Infer: adds 取值范围 and 使用条件 fields in addition to 公式 and 变量	中文名称, 英文名称, 子类别, 缩写符, 中文定义, 英文定义, 应用场景, 公式, 变量, 取值范围, 使用条件, 演示资源, 词性	[5 阶段] = '分析推断'	子类别, 编号 ID
Q06	Disciplinary	Q_学科_按子类别	Browse by Subcategory: wildcard parametric search on subcategory field; accepts fuzzy input (统计/抽样/推断/建模/实验/应用)	子类别, 中文名称, 英文名称, 缩写符, 中文定义, 应用场景, [5 阶段], 常用平台	子类别 LIKE '[input]*'	子类别, 编号 ID
Q07	Disciplinary	Q_学科_按学科领域	Browse by Discipline: wildcard search on the discipline field; accepts cross-disciplinary keywords (linguistics, psychology, ecology, economics, medicine, engineering)	学科领域, 中文名称, 英文名称, 缩写符, 中文定义, 应用场景, [5 阶段], 子类别	学科领域 LIKE '[input]*'	学科领域, 编号 ID
Q08	Disciplinary	Q_学科_跨学科汇总	Cross-disciplinary Aggregation: groups terms by discipline and pipeline stage, counting term frequency per cell; identifies most cross-disciplinary concepts	学科领域, COUNT(编号 ID) AS 术语数量, [5 阶段]	学科领域 IS NOT NULL	COUNT(编号 ID) DESC
Q09	Etymology	Q_词源_有词源记录	Terms with Etymology Records: retrieves all entries with a populated etymology field; supports browsing the lexical-historical dimension	词源, 中文名称, 英文名称, 中文定义, 英文定义, 子类别, [5 阶段]	词源 IS NOT NULL AND ≠ ''	词源, 编号 ID

Q ID	Category	Chinese Name	English Description	Fields Selected	Filter Condition	Sort Order
Q10	Etymology	Q_词源_含特定词根	Etymology Root Search: wildcard on etymology field; accepts Latin/Greek roots and eponymous names (Fisher, Pearson, Cohen, Bayes, Galton)	词源, 中文名称, 英文名称, 中文定义, 子类别, [5 阶段]	词源 LIKE '*[root]*'	词源, 编号 ID
Q11	Formula / Symbol	Q_公式_含公式术语	Terms with Formulas: retrieves all entries carrying a non-null formula field, together with variables, value ranges, definitions, and usage conditions	中文名称, 英文名称, 缩写符, 公式, 变量, 取值范围, 中文定义, 使用条件, 功能描述, 缺点, 子类别, [5 阶段]	公式 IS NOT NULL AND ≠ "	子类别, 编号 ID
Q12	Formula / Symbol	Q_公式_按符号查找	Symbol Reverse-lookup: wildcard across 公式 and 变量 simultaneously; user supplies symbol (σ , μ , χ , β , α , κ) and retrieves term	中文名称, 英文名称, 缩写符, 公式, 变量, 中文定义	(公式 OR 变量) LIKE '*[symbol]*'	子类别, 编号 ID
Q13	Lexical Contrast	Q_辨析_混淆词对照	Confusable Terms Browse (full): retrieves all entries with a populated 易混淆概念 field, displaying synonyms and antonyms alongside the target term	中文名称, 英文名称, 易混淆概念, 同义词, 反义词, 中文定义, 子类别, [5 阶段]	易混淆概念 IS NOT NULL AND ≠ "	子类别, 编号 ID
Q14	Lexical Contrast	Q_辨析_搜混淆词	Confusable Terms Parametric Search: wildcard across both the term name and the confusable-concepts field simultaneously	中文名称, 英文名称, 易混淆概念, 同义词, 反义词, 中文定义, 子类别, [5 阶段]	(中文名称 OR 易混淆概念) LIKE '*[input]*'	子类别, 编号 ID
Q15	Lexical Contrast	Q_辨析_按关键词全文检索	Full-text Keyword Search: simultaneous wildcard across	中文名称, 英文名称, 中文定义, 应用场景,	(name OR def OR 相关概念) LIKE	[5 阶段], 编号 ID

Q ID	Category	Chinese Name	English Description	Fields Selected	Filter Condition	Sort Order
			Chinese name, English name, Chinese definition, and related concepts; broadest-coverage discovery query	易混淆概念, 相关概念, [5 阶段], 子类别	'*[kw]*'	
Q16	COBUILD	Q_COBUILD_全部词条	All Entries in COBUILD Order: presents all 40 terms in corpus-lexicographic sequence (headword → POS → definition → example → collocation → pragmatics → media); no filter	英文名称, 中文名称, 词性, 缩写符, 英文定义, 中文定义, 例句 Prose, 例句来源, 术语搭配, 应用场景, 易混淆概念, 相关概念, 演示资源, 适用于可视化, 子类别, [5 阶段]	(none — all records)	[5 阶段], 编号 ID
Q17	COBUILD	Q_COBUILD_参数检索	COBUILD Parametric Search: wildcard across English name, Chinese name, prose example, and collocations simultaneously; corpus-driven discovery from usage context rather than term name	英文名称, 中文名称, 词性, 缩写符, 英文定义, 中文定义, 例句 Prose, 例句来源, 术语搭配, 应用场景, 易混淆概念, 相关概念, 演示资源, 适用于可视化, 子类别, [5 阶段]	(en_name OR zh_name OR 例句 OR 搭配) LIKE '[input]*'	[5 阶段], 编号 ID
Supp_Q1	COBUILD	Q_COBUILD_有演示资源	Entries with Visualization Resources: COBUILD-order retrieval filtered to entries with non-null 演示资源; sorted by suitability rating descending; functions as a navigation interface for the animation suite	英文名称, 中文名称, 词性, 缩写符, 适用于可视化, 演示资源, 英文定义, 中文定义, 例句 Prose, 例句来源, 术语搭配, 应用场景, 易混淆概念, 相关概念, 子类别, [5 阶段]	演示资源 IS NOT NULL AND ≠ "	适用于可视化 DESC, 编号 ID